Advanced DSP - NUIST

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

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Advanced DSP - NUIST (A35613)

Short Title:	Advanced DSP (NUIST)		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Science and Mathematics		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Postgraduate

Description of Module / Aims

The aim of this course is to build a deep understanding of DSP through the medium of MATLAB, no prior experience of DSP is assumed. The course addresses advanced topics relating to the modelling of digital systems, filter design, adaptive filter design and Wavelet Transforms. The course is focused on providing the student with necessary programming skills required to model different systems and observe their frequency response. The student is encouraged to use other DSP friendly programming languages such as Python to further develop their skills and understanding of this subject.

Programmes

`ADSP-0001`	MEng in Electronic Engineering (`WD_EELEN_R`)	1 / 1 / M
	MEng in Electronic Information Systems (`WD_EELEN_RI`)	1 / 1 / M
`ADSP-0001`	PG Dip in Engineering in Electronic Engineering (`WD_EELEN_G`)	1 / 1 / M

Indicative Content

- Z-Transform and difference equations
- Convolution, correlation and linear predictive coding
- Fourier transforms and filter design, quadrature mirror filters, haar transform
- Wavelet transforms and filter banks
- Wiener filters, Kalman gain and recursive least squres

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use MATLAB to determine the frequency response of a system from its Transfer function.
2. Design high pass, low pass and band pass / band stop filters using MATLAB.
3. Compute the coefficients and frequency response of filters using MATLAB.
4. Develop software to implement Weiner and self tuning adaptive filters.
5. Perform analysis using time-frequency representation, multi-level decomposition and basis functions.
6. Analyse and design sample rate changing systems and digital filterbanks.

Learning and Teaching Methods

- Lectures to lay down the foundations of theoretical grounding with integrated worked examples.
- Tutorials to develop problem solving skills
- Mini-project in a teamwork environment to develop simulation skills, handle complex tasks, integrate knowledge (acquired/self-acquired), evaluate and interpret results
- Virtual Learning Environment support (Moodle)

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,3,5,6
Final Project	40%	4

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

40% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	6	
Independent Learning	105	

Essential Material

- Leis, J.W. _Digital Signal Processing Using MATLAB for Students and Researchers_. UK: Wiley, 2011.
- Lyons, R.G. _Understanding Digital Signal Processing_. 3rd ed.. UK: Prentice Hall, 2011.

Supplementary Material

- Deller, J.R., J.G. Proakis and J.H.L. Hansen. _Discrete time processing of speech processing_. -: Wiley, 2000 - Seminal.
- Madisetti, V.K. and D.B. Williams. _Digital Signal Processing Handbook_. .: CRC Press, 1999.